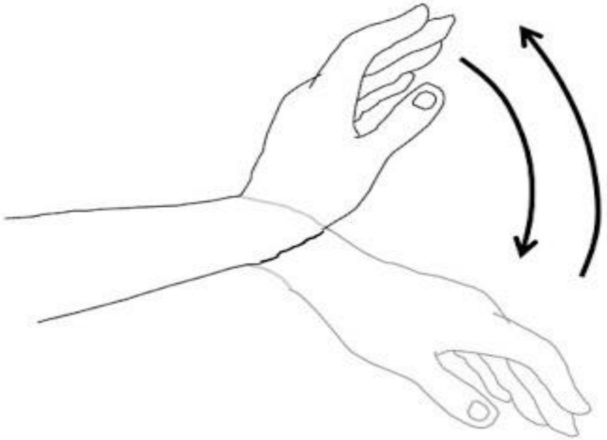
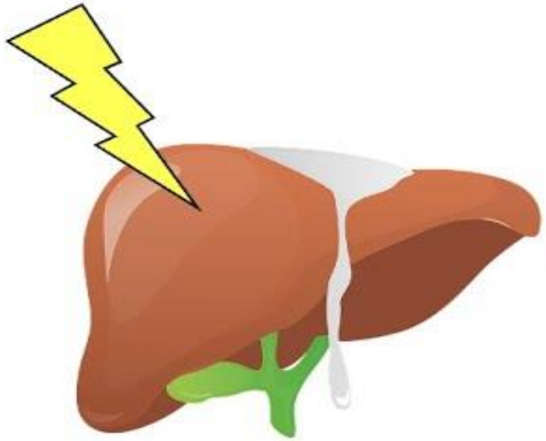
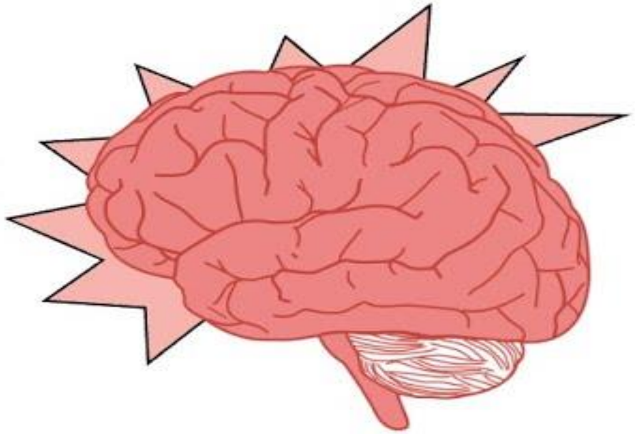
Hepatic Encephalopathy

Prepared by

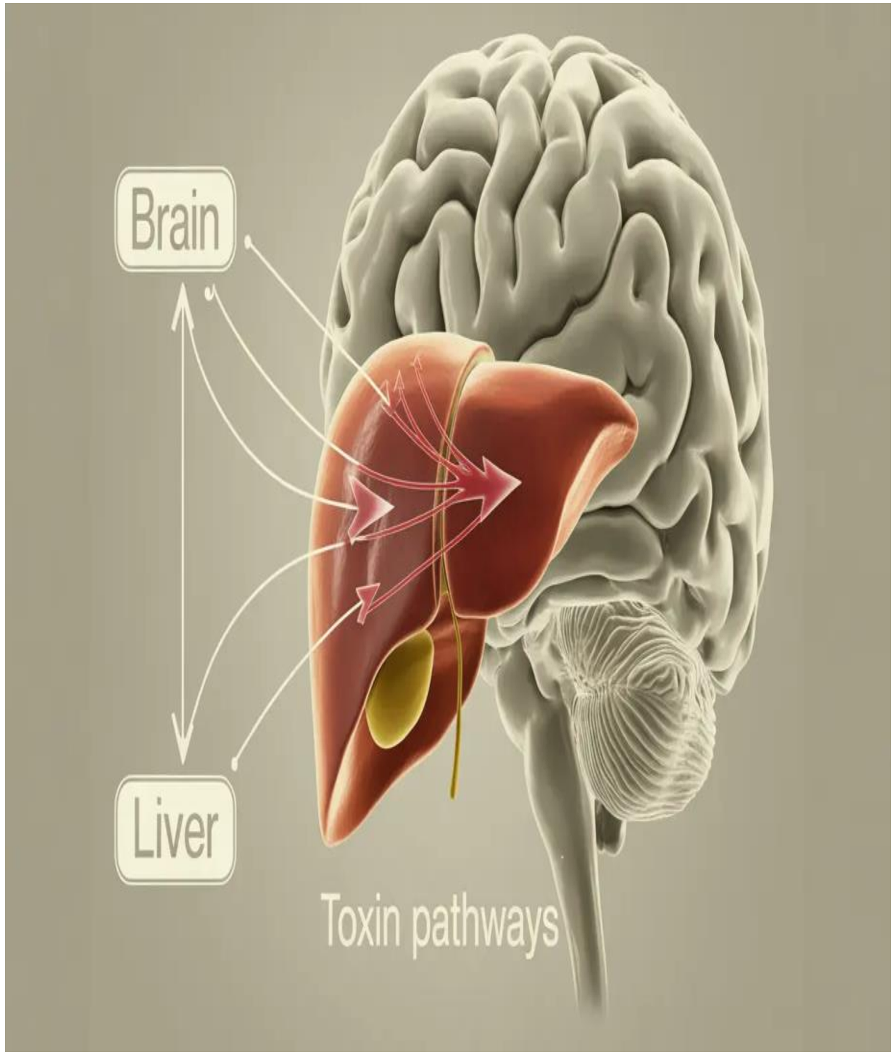
Assist professor Dr \Afaf Mohamed Mansour

Outlines:

- Description
- Pathophysiology
- Source of Ammonia
- Others cause of Hepatic Encephalopathy
- Stage of Hepatic Encephalopathy
- Clinical Manifestations
- Assessment and Diagnostic Findings
- Management of Hepatic Encephalopathy

Hepatic Encephalopathy

Hepatic Encephalopathy is a life-threatening complication of liver disease that occurs with profound liver failure. It is the neuropsychiatric manifestation of hepatic failure associated with portal hypertension and the shunting of blood from the portal venous system into the systemic circulation.



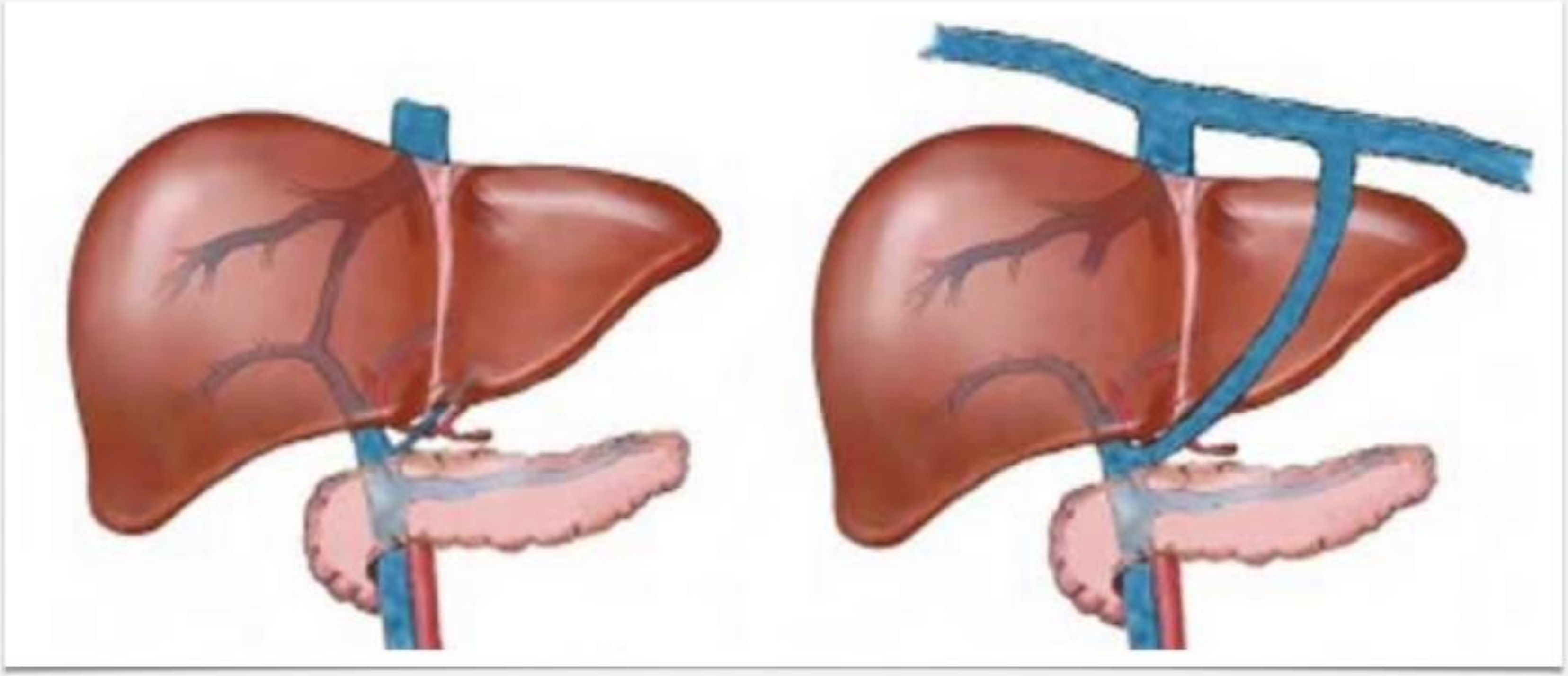
Causes of hepatic encephalopathy

Two major alterations underlie its development in acute and chronic liver disease includes **hepatic insufficiency and portal systemic shunting.**

Hepatic Insufficiency may result in encephalopathy because of the inability of the liver to detoxify toxic by products of metabolism.

Portal Systemic Shunting in which collateral vessels develop as a result of portal hypertension, allows elements of the portal blood (laden with potentially toxic substances usually extracted by the liver) to enter the systemic circulation.

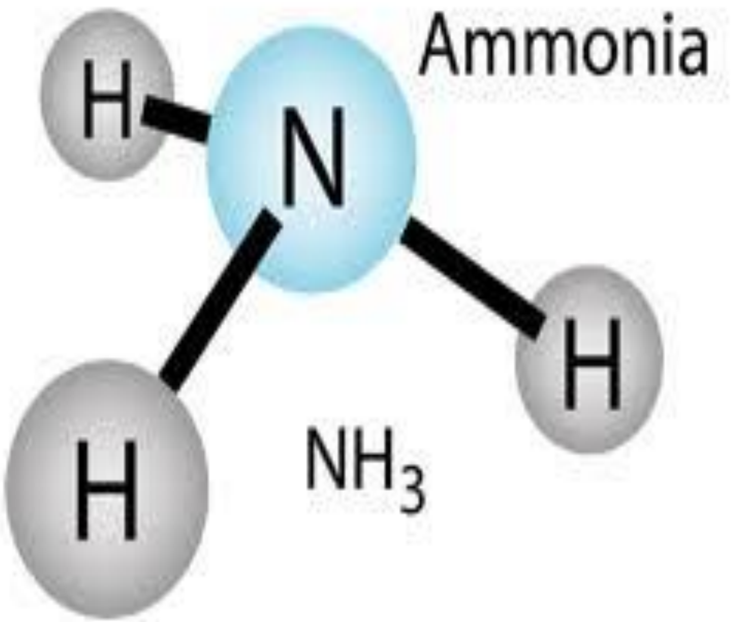
PORTOSYSTEMIC SHUNTS



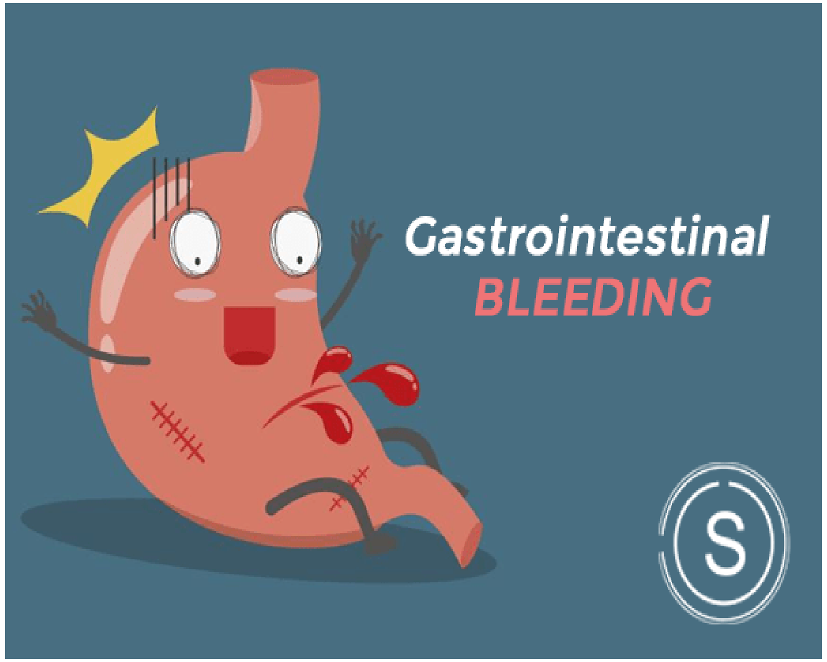
Normal Liver *Portal Hypertension*

Causes of hepatic encephalopathy

Ammonia is considered the major etiologic factor in the development of encephalopathy. It enters the brain and excites peripheral receptors on astrocyte cells causing depression of the central nervous system. Ammonia inhibits neurotransmission and synaptic regulation, producing sleep and behavior patterns change associated with hepatic encephalopathy.



Source of Ammonia

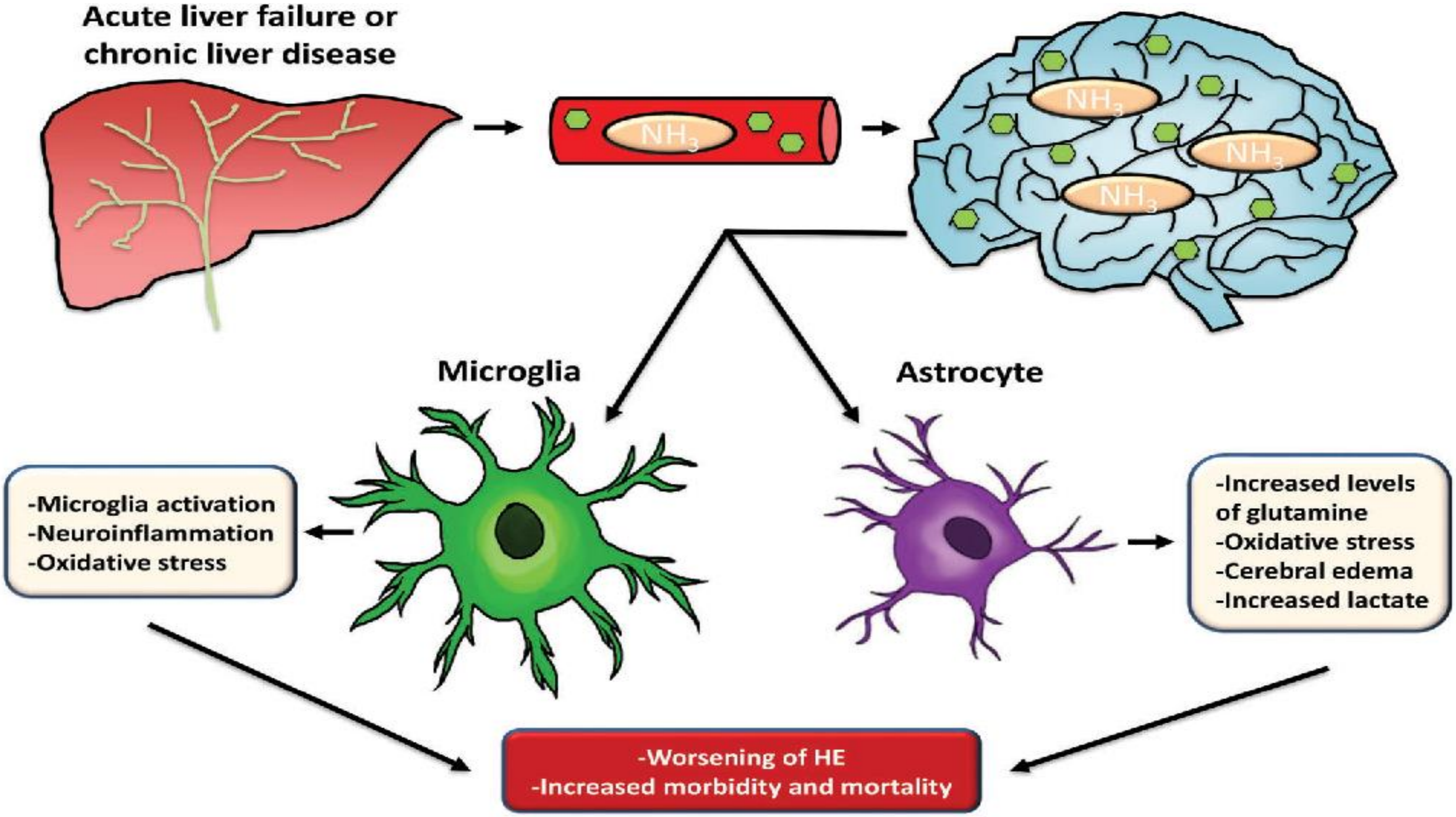


- ❖ GI bleeding (bleeding esophageal varices, chronic GI bleeding).
- ❖ A high-protein diet.
- ❖ Bacterial infection or uremia.
- ❖ The ingestion of ammonium salts.
- ❖ Presence of alkalosis or hypokalemia, increased amounts of ammonia is absorbed from the GI tract and from the renal tubular fluid.

Others cause of Hepatic Encephalopathy:

- ✓ Excessive diuresis
- ✓ Dehydration, infections
- ✓ Surgery
- ✓ Fever
- ✓ Some medications (sedatives, tranquilizers, analgesics, and diuretics that cause potassium loss).

Pathophysiology of hepatic encephalopathy



Stages of Hepatic Encephalopathy

Stage One:

- Normal level of consciousness with periods of lethargy
- Euphoria; reversal of day–night sleep patterns.
- Asterixis; impaired writing and ability to draw line figures
- Normal EEG.

Stages of Hepatic Encephalopathy

Stage Two:

- Increased drowsiness , disorientation
- inappropriate behavior
- mood swings; agitation.
- Asterixis
- Fetor hepaticus
- Abnormal EEG with generalized slowing.

Stages of Hepatic Encephalopathy

Stage Three:

- Stuporous; difficult to arouse
- Sleeps most of time
- Marked confusion
- Incoherent speech.
- Asterixis, Increased deep tendon reflexes; Rigidity of extremities
- Markedly abnormal EEG.

Stages of Hepatic Encephalopathy

Stage Four:

- Comatose; may not respond to painful stimuli.
- Absence of Asterixis
- Absence of deep tendon reflexes
- Flaccidity of extremities
- Markedly abnormal EEG.

Clinical Manifestations

- **Minor mental changes and motor disturbances.**

The patient appears slightly confused and has alterations in mood and sleep patterns. The patient tends to sleep during the day and has restlessness and insomnia at night.

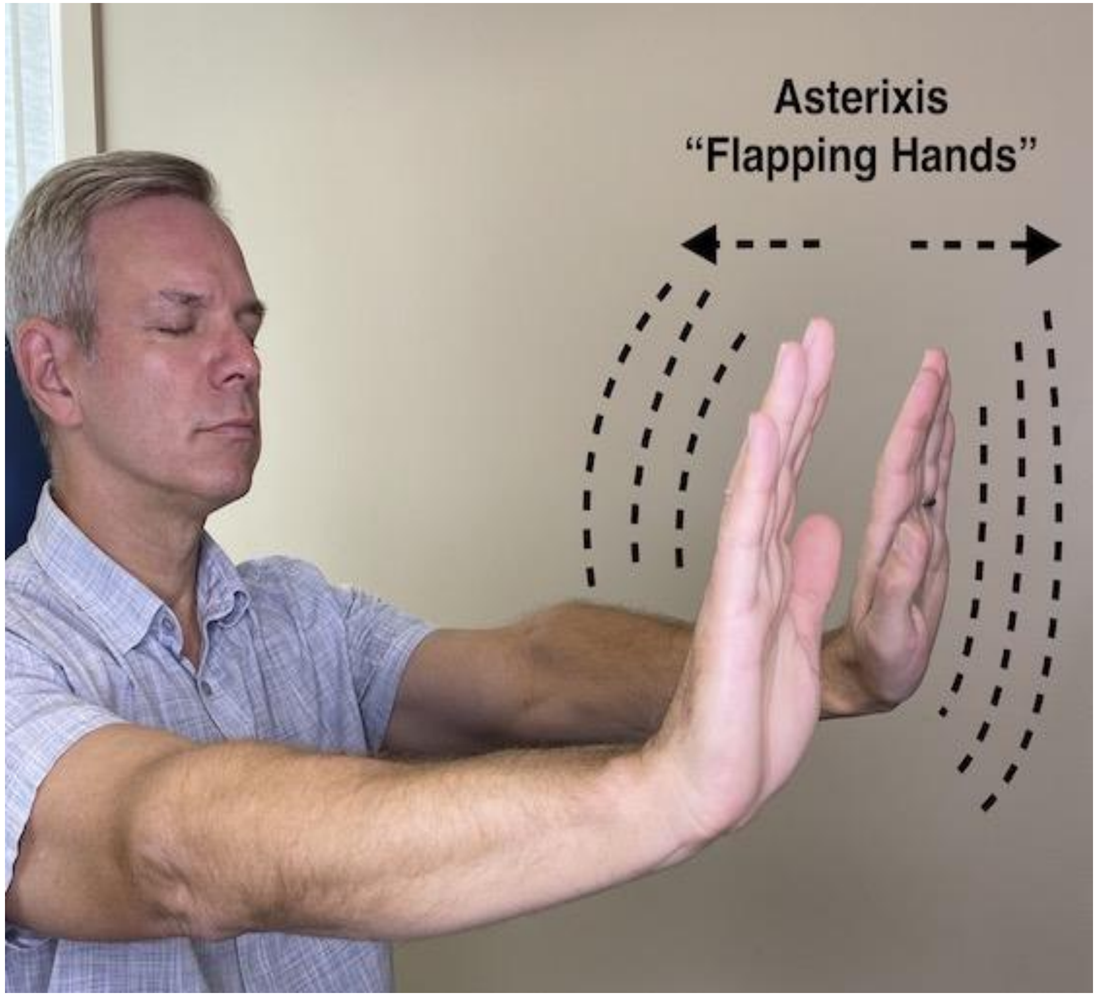
- As hepatic encephalopathy progresses, the patient may become difficult to awaken and completely disoriented with respect to time and place. With further progression, the patient lapses into frank coma and may have seizures.

Clinical Manifestations

* **Asterixis** (flapping tremor of the hands) may be seen in stage II Encephalopathy.

Asterixis Simple tasks, such as handwriting, become difficult. A handwriting or drawing sample (star figure)

* Fetor Hepaticus (*a sweet, slightly fecal odor* to the breath that is presumed to be of intestinal origin) may be noticed.



Assessment and Diagnostic Findings

The electroencephalogram (EEG) shows a generalized slowing, an increase in the amplitude of brain waves and characteristic triphasic waves.

Management of Hepatic Encephalopathy

- Initiating ammonia-lowering therapy such as:

Lactulose: It acts by several mechanisms that promote the excretion of ammonia in the stool: these mechanisms are:

(1) Ammonia is kept in the ionized state, resulting in a decrease in colon pH, reveres in the normal passage of ammonia from the colon to the blood.

(2) Evacuation of the bowel takes place, which decreases the ammonia absorbed from the colon.

(3) The fecal flora is changed to organisms that do not produce ammonia from urea.

* The patient receiving lactulose is monitored closely for the development of watery diarrheal stools, because they indicate a medication overdose.

Lactulose:

Side effects of Lactulose include intestinal bloating and cramps, which usually disappear within a week. To mask the sweet taste, which some patients dislike, it can be diluted with fruit juice.

Nursing Consideration: The patient is closely monitored for hypokalemia and dehydration. Other laxatives are not prescribed during lactulose administration because their effects disturb dosage regulation.

Route: Lactulose may be administered by nasogastric tube or enema for patients who are comatose or for those in whom oral administration is contraindicated or impossible.

Initiating ammonia-lowering therapy such as:

Antibiotics as Neomycin: Metronidazole (Flagyl), and Rifaximin (Xifaxan) have been used to reduce levels of ammonia-forming bacteria in the colon.

- ***Action:*** is broad – spectrum antibiotic that destroy, normal bacteria found in bowel, thereby decreasing protein break down and ammonia production.
- ***Route:*** Neomycin is given orally every 4 to 6 hours.
- ***Side effect:*** This drug is toxic to the kidney and cannot be given to patients with renal failure.
- ***Nursing Consideration***: Daily renal function studies are monitored when Neomycin is administered.

Additional Principles of Management of Hepatic Encephalopathy

- Assessing neurological status frequently.
- Monitoring mental status by keeping a daily record of handwriting and arithmetic performance.
- Recording intake and output and body weight daily.
- Measuring and recording vital signs every 4 hours.
- Potential sites of infection (peritoneum, lungs) are assessed frequently, and abnormal findings are reported promptly.

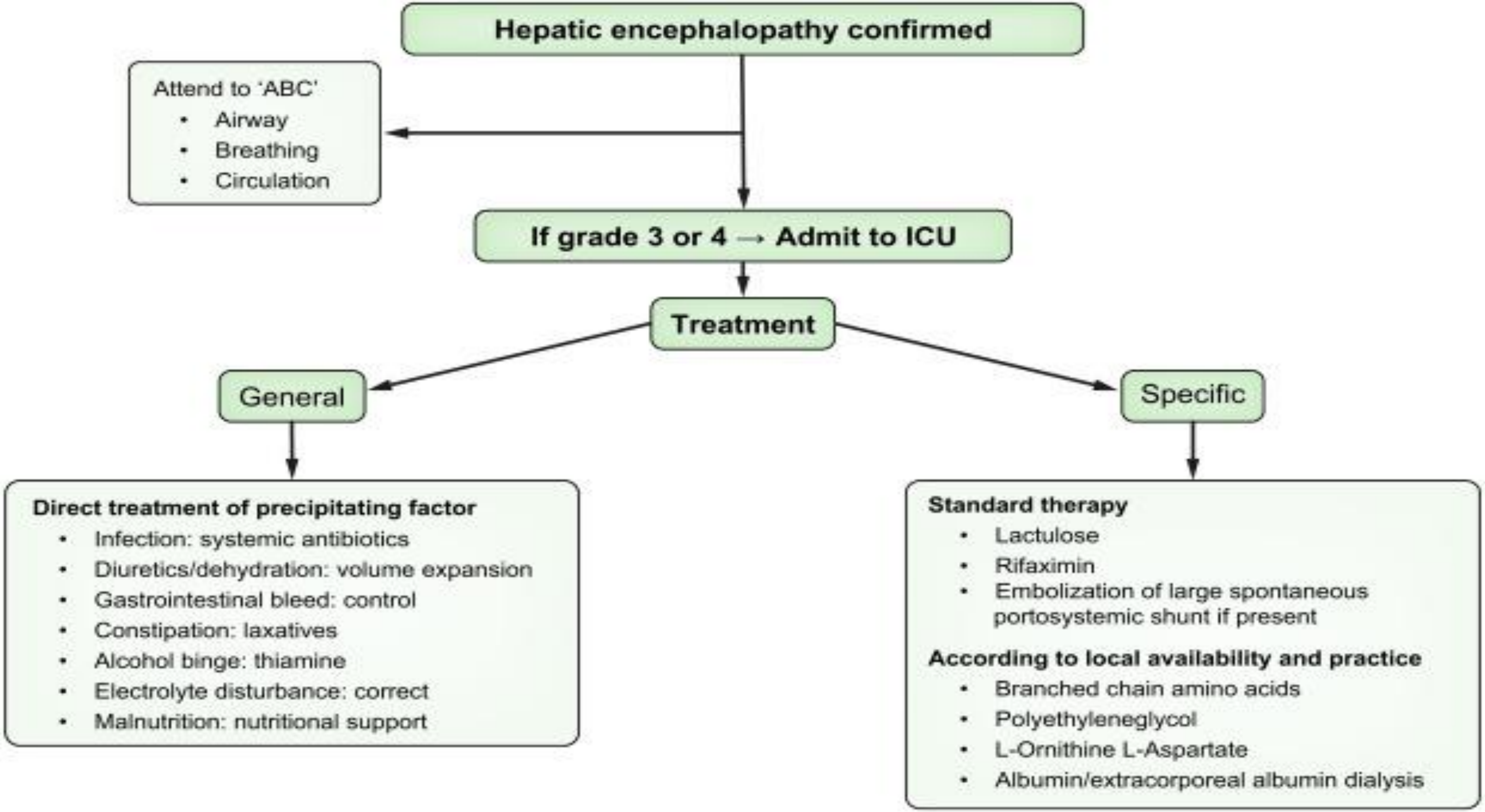
Additional Principles of Management of Hepatic Encephalopathy

- Serum ammonia level is monitored daily.
- Protein intake is moderately restricted in patients who are comatose or who have encephalopathy that is refractory to lactulose and antibiotic therapy.

- ***Long-term restriction of dietary protein to less than 1 g/kg daily should be avoided.***

- If animal protein precipitates encephalopathy, vegetable or dairy proteins may be used as most patients can tolerate a diet of vegetable protein up to 120 g/day.

- Patients and families are advised about foods that are high in protein (meat, eggs), which may need to be eliminated from the diet for short term to reduce production of ammonia.


Hepatic encephalopathy confirmed
Attend to 'ABC'
• Airway
• Breathing
• Circulation
If grade 3 or 4 → Admit to ICU
Treatment
General
Specific
Direct treatment of precipitating factor
• Infection: systemic antibiotics
• Diuretics/dehydration: volume expansion
• Gastrointestinal bleed: control
• Constipation: laxatives
• Alcohol binge: thiamine
• Electrolyte disturbance: correct
• Malnutrition: nutritional support
Standard therapy
• Lactulose
• Rifaximin
• Embolization of large spontaneous portosystemic shunt if present
According to local availability and practice
• Branched chain amino acids
• Polyethyleneglycol
• L-Ornithine L-Aspartate
• Albumin/extracorporeal albumin dialysis

Nutritional Management

- Prevent the formation and absorption of toxins, principally ammonia, from the intestine.
- Keep daily protein intake between 1.0 and 1.5 g/kg, depending on the degree of decompensation.

  *Avoid protein restriction if possible, even in those with encephalopathy. If necessary, implement temporary restriction of 0.5 to 0.8 g/kg.

- For patients who are truly protein-intolerant, provide additional nitrogen in the form of an amino acid supplement.

Nursing Management

* Maintaining a safe environment to prevent injury, bleeding and infection.
* The nurse administers the prescribed treatments and monitors the patient for the numerous potential complications.
* The nurse encourages deep breathing and position changes to prevent the development of atelectasis, pneumonia and other respiratory complications.
* Despite aggressive pulmonary care, patients may develop respiratory compromise. They may require intubation and mechanical ventilation to protect the airway and they are frequently admitted to the ICU

Teaching Patients Self Care

- If the patient has recovered from hepatic encephalopathy and discharged home, the nurse instructs the family to watch for slight signs of recurrent encephalopathy.
- In the acute phase of hepatic encephalopathy, dietary protein may be reduced for a brief period to 0.8 to 1.0 g/kg per day.
- During recovery and in the home situation, it is important to the patient in maintenance of a moderate protein, high calorie diet.
- Protein may then be added in 10 g increments every 3 to 5 days if mental status is improving.


Thank you